

Inequalities – 4

Inequalities for series and sequences

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Theoretical minimum.

Theorem 1 (Abel summation). With $A_k = a_1 + \dots + a_k$ one has $\sum_{k=1}^n a_k b_k = A_n b_n + \sum_{k=1}^{n-1} A_k (b_k - b_{k+1})$. If b_k is monotone and the partial sums A_k are bounded by M , then $|\sum a_k b_k| \leq M(|b_1| + 2|b_n|)$. This is the discrete second mean value theorem and the engine behind the Dirichlet and Abel tests.

Theorem 2 (Hardy and Carleman). For $a_k \geq 0$ and $p > 1$, $\sum_{n \geq 1} \left(\frac{a_1 + \dots + a_n}{n} \right)^p \leq \left(\frac{p}{p-1} \right)^p \sum a_n^p$, and the constant is sharp. Letting $p \rightarrow \infty$ after a substitution gives Carleman's inequality $\sum_{n \geq 1} (a_1 \dots a_n)^{1/n} \leq e \sum a_n$, again with a sharp constant.

Fact 1 (Hilbert). $\sum_{m,n \geq 1} \frac{a_m b_n}{m+n} \leq \pi (\sum a_m^2)^{1/2} (\sum b_n^2)^{1/2}$, with π sharp. All three theorems are proved the same way: split the term, apply Hölder with a weight, and choose the weight so that the resulting constant is finite.

1. Prove Abel summation and the bound following it, and deduce the Dirichlet and Abel tests for convergence of a series.
2. Prove Hardy's inequality for a general $p > 1$ with the sharp constant $\left(\frac{p}{p-1} \right)^p$ — the case $p = 2$ was done in Differentiable functions – 9 — and deduce Carleman's inequality with the constant e by letting $p \rightarrow \infty$ after a substitution.
3. (VJIMC 1994) Prove that $\prod_{i=1}^n \left(1 + \frac{1}{2^i} \right) < 3$ for every n .
4. (VJIMC 2001) Let $n \geq 2$. Prove that $\prod_{k=2}^n \ln k < \frac{\sqrt{n!}}{n}$.
5. (VJIMC 2022) Let $a_1 > 0$ and $a_{n+1} = a_n + \frac{1}{a_1 + a_2 + \dots + a_n}$ for $n \geq 1$. Prove that $\lim_{n \rightarrow \infty} \frac{a_n^2}{\ln n} = 2$.